Environment

Get global view through the SDN controller

SDN global views

Task information

Model inputs

Experience Replay Pool

$(s_t, a_t, R_t, s_{t+1}, P_t)$

Global Network

Actor Network

s_t

a_t

Critic Network

s_t

$V_{\theta_v}(s_t)$

Model inputs

s_t

Update model parameters

Reward Model

State Encoder

Action Encoder

Fusion (Concat)

Reward Head

R_t

Calculation of the advantage function

Upload gradient

Synchronization parameter

Upload gradient

Synchronization parameter

Worker 1

Actor Network

Critic Network

Experience Replay Pool

Worker n

Actor Network

Critic Network

Experience Replay Pool

Extract experience

Extract experience

Environment

Get global view through the SDN controller

SDN global views

Task information

Model inputs

s_t

Environment input flow (SDN views / Task info)

Actor/Critic flow (Policy & Value)

Reward / Advantage flow (Scalar reward R_t)

Worker ↔ Global sync flow (Upload/Synchronization)